# Residual Corrected Flow for Human-Centric Editing Across Video Shots and 3D Assets

Wenshuo Gao, Jiangyue Zeng, and Shuai Yang, *Member, IEEE,*

***Abstract*—Human-centric editing seeks to change scenes, clothing, props, and illumination while preserving the subject's identity and performance. Conditional generation and direct editing can entangle the intended changes with source reconstruction errors, whereas strong source constraints can inhibit the requested edits. We present FlowCine, a training-free framework for video and 3D editing grounded in condition consistency: unchanged conditions should reconstruct the source, making reconstruction the reference for controlled editing. Our residual-corrected flow transfers the correction required for source reconstruction to target generation, anchoring the edited trajectory to the observed input. To reconcile preservation with editability, we decouple appearance from shared structural conditions and localize residual correction and frequency transfer to identity-bearing regions. This preserves facial details and recorded motion while allowing the requested appearance changes under text instructions and independently optional visual references. We extend this formulation to sparse 3D shape and PBR appearance latents, where spatial masks and coordinate-aware frequency transfer preserve identity in the asset's coordinates during geometry and material editing. Experiments on video and 3D editing demonstrate improvements over the compared methods in source preservation, edit adherence, and visual quality. Project Page: https://gaowenshuo.github.io/FlowCine/.**

***Index Terms*—Video editing, flow matching, 3D editing.**

## I. Introduction

BRINGING a visual concept to life through physical production is costly. Securing locations, preparing costumes and props, and arranging lighting require substantial resources and coordination, with each visual design imposing its own production demands. Transforming these elements in modestly budgeted footage to meet higher visual and production standards would allow a single recorded performance to support diverse creative visions. This capability has broad applications in filmmaking, virtual reality, advertising, and creative media, while also paving the way for future world-to-world applications. Human-centric editing offers this flexibility by changing the scene, clothing, objects, and illumination while retaining the performer's identity, details, and motion. For 3D human assets, similar control would make these visual choices adjustable across viewpoints, supporting the reuse of digital performers in virtual production.

Jointly changing setting, clothing, and objects creates a conflict between preservation and editability. Directly applying these edits can change the person's face and fine details, weakening identity consistency. Conversely, strong constraints on the source appearance can preserve unwanted clothing textures, object shapes, or background content, preventing the requested changes. This trade-off becomes more challenging in video, where both the preserved details and the edited appearance must remain consistent as the person moves. An effective editing framework must protect identity-related information while allowing the regions involved in the edit to change freely according to the request.

We present **FlowCine**, a training-free framework for controllable human-centric editing of videos and 3D assets, illustrated in Fig. 1. Users specify an edit in natural language and may add background, clothing, and object images independently. A pretrained image editor combines these conditions into a target appearance, and a pretrained video/3D generative model transfers that appearance while following the source structure. This separates the choice of visual design from the recorded performance and provides the same interface for text-driven and reference-guided edits.

Specifically, our key insight is to exploit the principle of condition consistency that every change in the output is driven by a change in the target condition (*i.e.*, illumination). It implies a perfect reconstruction when there is no change in condition. Based on this principle, we introduce a residual-corrected flow mechanism that rephrases a typical flow-based generation model to a novel training-free editing model. It ensures structural consistency by directing the flow towards perfect reconstruction when the target and source conditions are identical, so that subsequent changes in the conditions like illumination variation can be reflected in the video output.

For video editing, we propose decoupled conditions and identity-aware regional structural control to separate lighting from structural guidance. Specifically, we introduce region-specific designs for the head, other human foreground regions, and the background, which enhance the model's adherence to conditional guidance. To further preserve fine-grained details in the head region, we introduce frequency transfer, which explicitly transfers high-frequency details from the source video to the edited output. Together, these components balance identity preservation, detail consistency, and editing flexibility within one flow-based process.

For 3D editing, FlowCine encodes a supplied asset into shape and physically based rendering (PBR) appearance latents. Multi-view semantic labels define a spatial head region, and residual correction operates directly in both latent spaces. Coordinate-aware frequency transfer uses the XYZ locations of sparse features to preserve local detail. The edited latents are decoded into an asset that can be viewed from new directions.

This work extends our previous work, FlowPortal [1],

Wenshuo Gao, Jiangyue Zeng, and Shuai Yang are with the Wangxuan Institute of Computer Technology, Peking University, Beijing 100080, China (e-mail: {gaowenshuo, zengjy}@stu.pku.edu.cn; williamyang@pku.edu.cn).
Corresponding author: Shuai Yang.

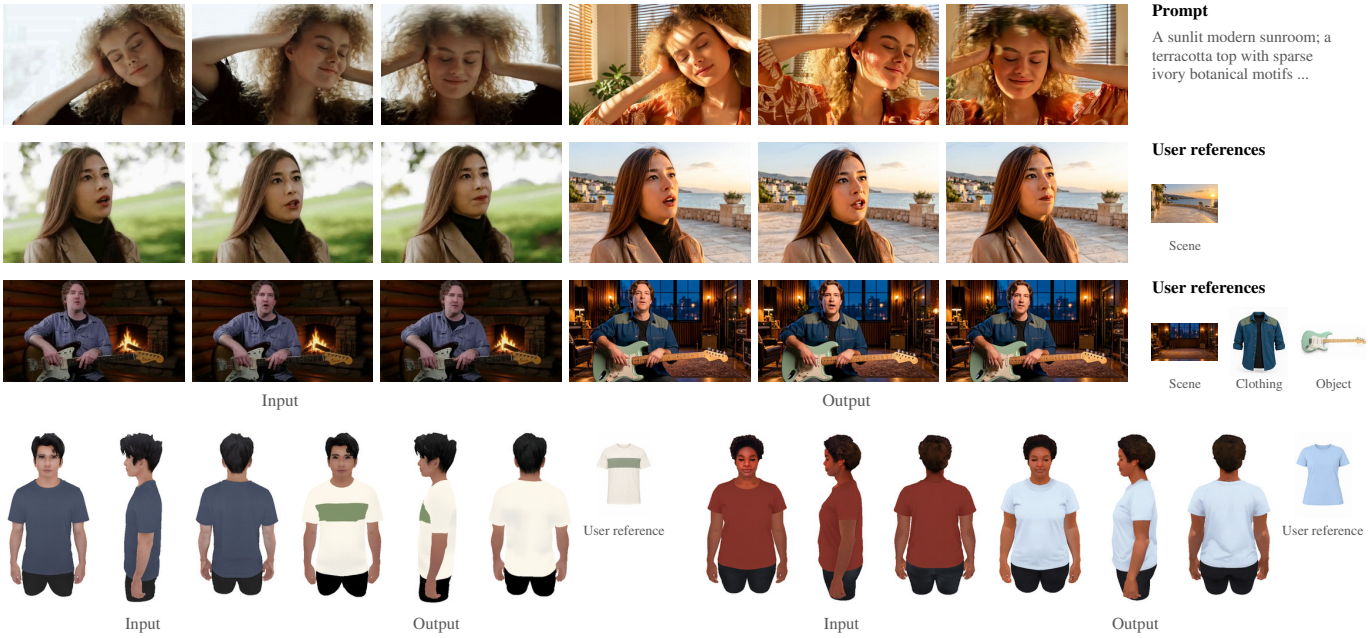



Fig. 1. Human-centric video and 3D editing. FlowCine supports text-guided and reference-guided changes to scenes, clothing, objects, and illumination while preserving human identity, details, and motion. Users can independently supply scene, clothing, and object references, and edit 3D assets with consistent appearance across viewpoints. Small thumbnails display the provided reference images, while brief prompt summaries are shown for text-only edits.

which studied video relighting and background replacement. FlowCine broadens the task to joint scene, clothing, prop, and illumination editing with independently optional visual references. Its technical additions include region-dependent structural control and head preservation for video, and 3D shape and PBR appearance editing with semantic spatial masks and coordinate-aware frequency transfer. We also expand the experimental evaluation with video and multi-view 3D comparisons, module ablations, and regional structure, detail, and motion metrics. In summary, our contributions are threefold:

- A unified human-centric editing framework covering scene, clothing, prop, and illumination changes, with independently optional visual references and a shared interface for video and 3D assets.
- A condition consistency formulation and a residual-corrected flow method that transfers source reconstruction corrections to target generation to preserve source content while enabling the requested edits.
- Region-dependent structural control and detail transfer for separating identity-related information from editable appearance, together with coordinate-aware residual correction and frequency transfer in sparse 3D latents.

The remainder of the paper is organized as follows. Section II reviews related work. Sections III and IV present residual-corrected flow for human-centric video editing and its extension to 3D assets, respectively. Section V reports comparisons and ablation studies. We discuss limitations and conclude in Sections VI and VII.

## II. Related Work

### A. Human-Centric Video Editing

Video editing balances requested changes against source preservation. Inversion-based approaches [2]–[5] connect editing to reconstruction, while FateZero [6], TokenFlow [7], and FRESCO [8] use attention or correspondence constraints for consistent propagation. AnyV2V [9] propagates an image-edited first frame through temporal feature injection, FastVideoEdit [10] uses consistency models for efficient sampling, and FlowEdit [11] constructs inversion-free trajectories with pretrained flow models.

VideoPainter [12] introduces plug-and-play masked-video context control, while VACE [13] unifies source videos, masks, and visual references within one conditioning interface. Kiwi-Edit [14], Lucy Edit 1.1 [15], UniVideo [16], and DITTO [17] train instruction-guided editing, while Kiwi-Edit and UniVideo also accept reference images. Omni-Video 2 [18] integrates multimodal understanding, generation, and editing. VIVA [19] uses vision-language conditioning and reward-based post-training, while EasyV2V [20] studies data construction and flexible input conditioning. Qwen-Video-Edit [21] adapt image editors to video latents, and VicEdit [22] learns from image, image-pair, and video-pair editing examples alongside text. EditVid [23] combines temporal memory, token injection, and latent blending for training-free text- and reference-guided editing. These methods face a trade-off between editing flexibility and detail preservation. Direct editing often struggles to preserve fine source details, while stronger source constraints can restrict the requested appearance changes.

### B. Video Relighting and Background Replacement

Video relighting preserves surface appearance, geometry, and motion while changing illumination. The Relightables [24] enables relightable volumetric performance capture, while Neural Video Portrait Relighting [25] achieves real-time portrait relighting through temporal consistency modeling. Total Relighting [26] and SwitchLight [27] estimate normals

and albedo for lighting control, while Relightful Harmonization [28] and IC-Light [29] train foreground relighting and illumination harmonization with supplied backgrounds.

RelightVid [30] and Lux Post Facto [31] train video diffusion models for consistent relighting, including portrait performances. TC-Light [32] and Lumen [33] train background replacement on long videos. IllumiCraft [34] and UniRelight [35] couple illumination with geometry and intrinsic decomposition, respectively. LumiSculpt [36] studies consistent portrait lighting, and UniLumos [37] unifies image and video relighting with physics-plausible feedback. However, collecting large-scale high-quality paired video relighting datasets is difficult and render-based datasets often lack lighting richness. As a result, they struggle to balance between the lighting conditions and the video foreground conditions. Therefore, the trained models tend to produce lighting results with insufficient richness and diversity in illumination, and lacking fidelity under complex conditions.

Training-free AnyPortal [38] and Light-A-Video [39] combine IC-Light with video models through cross-frame attention with refinement projection, and consistent light attention with progressive light fusion, respectively. However, they use IC-Light to relight per frame individually, introducing frame discontinuities that cannot be fully corrected by the video model. In addition, due to the lack of sufficient control in the video model, the relighting results may exhibit inconsistencies in structure and motion compared to the original video. Over-complicated pipelines also make them less inference-efficient.

Our method is also training-free. By introducing the proposed residual-corrected flow and frequency transfer, we better address structural and detail consistency. To achieve stronger conditional control, we propose decoupled conditions and identity-aware regional structural control to enhance the model's ability to respond to given conditions. We also leverage residual reusing to achieve effective inference.

### C. Appearance Editing and 3D Priors

Appearance editing modifies object or scene appearance while preserving geometric structure and multi-view consistency. Text2Tex [40] and TexFusion [41] use image diffusion for mesh texturing through depth conditioning and shared multi-view texture denoising. Instruct-NeRF2NeRF [42] alternates image editing with radiance-field optimization, while GaussianEditor [43] uses semantic tracing and hierarchical Gaussian representations for localized scene editing.

TRELLIS [44] and TRELLIS.2 [45] provide structured latents for geometry and appearance, with rectified-flow generation, compact geometry, and material representations. AnchorFlow [46] aligns source and target flow trajectories with latent anchors for training-free shape editing, and 3D-LATTE [47] performs training-free geometry editing in 3D diffusion latents. Easy3E [48] combines voxel flow editing with normal-guided multi-view texture refinement. Hunyuan3D 2.1 [49] paints PBR materials on supplied meshes, while Steer3D [50] learns text-controlled image-to-3D editing from synthetic data. Vinedresser3D [51] and Prox-E [52] localize edits through agent-guided latent inpainting and primitive-based shape abstractions, translating instructions into spatial guidance for generative models. PriorEdit3D [53] learns feed-forward editing without paired 3D supervision by combining image-editing, semantic, and 3D priors.

However, image-based editing pipelines often suffer from cross-view inconsistencies, while existing latent-space 3D editing methods struggle to preserve identity-sensitive structures and fine-grained details. To address these limitations, we propose a unified training-free framework that integrates residual-corrected flow into both shape and PBR appearance latents for consistent 3D editing. We further introduce identity-aware regional control based on multi-view semantic labels and coordinate-aware Frequency Transfer to preserve fine-grained details through 3D spatial correspondence.

## III. Residual-Corrected Flow for Human-Centric Video Editing

### A. Preliminaries

FlowCine edits a source video or a textured human asset according to a text instruction and optional visual references. Our aim is to change the requested appearance while preserving the subject's identity and source structure, motion, and details. Pretrained conditional flow models supply the generative priors in the corresponding latent spaces.

*1) Conditional flow models:* Let $z_t$ denote a latent at time $t$, where $t = 1$ corresponds to Gaussian noise and $t = 0$ to a clean representation. Under condition $c$, the model predicts the denoising velocity

$$V_t^c(z_t) = F_\theta(z_t, t, c), \qquad \frac{\mathrm{d}z_t}{\mathrm{d}t} = -V_t^c(z_t). \tag{1}$$

Generation follows this field from $t = 1$ to $t = 0$. In practice, we discretize the time interval into $N$ steps using $N+1$ time points $0 = t_0 < t_1 < \cdots < t_N = 1$. Starting from $z_{t_N} = \epsilon \sim \mathcal{N}(0, I)$, an Euler step gives

$$z_{t_{i-1}} = z_{t_i} + (t_i - t_{i-1}) V_{t_i}^c(z_{t_i}), \tag{2}$$

for $i = N, \ldots, 1$. The final latent $z_{t_0}$ is decoded into the generated output. We use this denoising convention for both video and 3D editing.

*2) Sparse 3D latent representations:* The 3D branch uses the sparse shape and physically based rendering (PBR) appearance representations of TRELLIS.2 [45]. A textured asset is encoded into shape features $z_0^{\mathrm{sh}}$ and appearance features $z_0^{\mathrm{tex}}$ on a common set of spatial coordinates $\mathcal{C}$. Each feature row is associated with an XYZ location in $\mathcal{C}$. The shape features encode geometry, while the appearance features encode material properties for PBR. Separate image-conditioned flow models operate on the two feature spaces. Shape is generated first, and the resulting shape features additionally condition appearance generation. The corresponding decoders convert these latent representations into a textured 3D asset that can be rendered from different viewpoints. In the following, $z_t$ and $F_\theta$ denote the latent and flow model of the relevant modality or stage, unless specified otherwise.

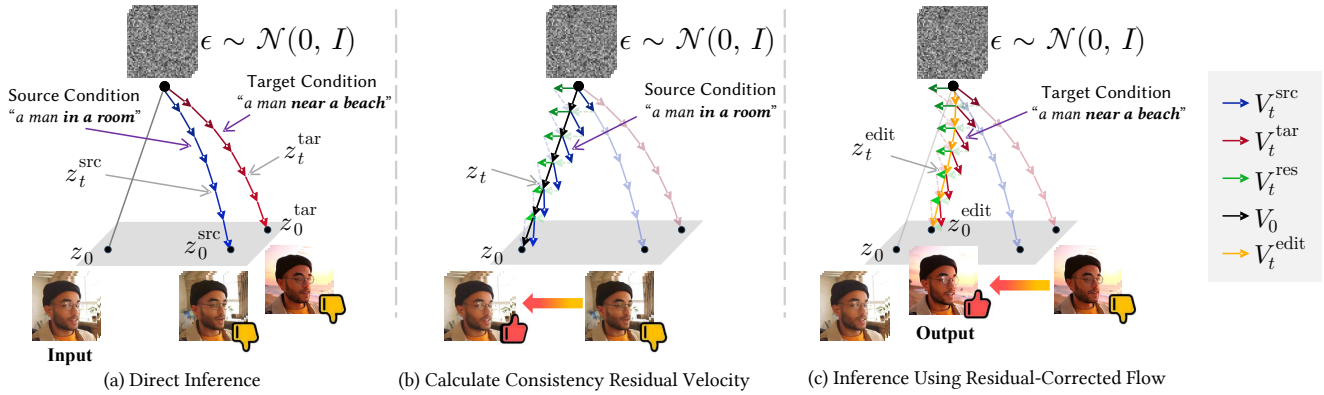



Fig. 2. **Illustration of Residual-Corrected Flow.** (a) Direct inference builds source- and target-conditioned denoising trajectories from the same noise $\epsilon$. The generated source $z_0^{\text{src}}$ differs from the recorded input $z_0$. (b) The consistency residual velocity $V_t^{\text{res}}$ is the difference between the input reconstruction velocity $V_0$ and the predicted source velocity $V_t^{\text{src}}$, correcting the source trajectory toward the input. (c) Combining the target velocity $V_t^{\text{tar}}$ with this residual produces the edited trajectory, preserving source structure while following the target condition. Purple arrows indicate the conditioning used to predict velocities. For simplicity, we omit the conditions other than prompt.

### *B. Condition Consistency*

**Condition consistency** expresses the relationship between a requested edit and the resulting change in the editing process. An ideal editor responds to the difference between source and target conditions while retaining attributes indentical that the instruction leaves unchanged. We formulate this requirement through two properties.

*1) Directional change:* When the source and target conditions differ in a particular attribute, the output should change in that attribute while preserving the remaining content. For example, when the source and target conditions differ only in lighting, the output should change accordingly while preserving structure, motion, and fine details unrelated to illumination.

*2) Stability under identity:* In the extreme case where the source and target conditions are identical, an ideal editing process should reconstruct the input. A zero condition difference should therefore produce a zero edit. Faithful source reconstruction provides the reference from which a controlled edit can be defined.

Direct conditional generation exposes the connection between these properties. As illustrated in Fig. 2(a), one can start from the same noise and generate $z_0^{\text{src}}$ and $z_0^{\text{tar}}$ under the source and target conditions. The condition describes the input incompletely, and the pretrained model introduces its own reconstruction error. As a result, the source-conditioned output can differ significantly from the input latent $z_0$. Consequently, editing a real recording combines the intended condition change with this source reconstruction discrepancy. Even when source and target conditions are identical, the editing result fails to produce the input.

Our approach first corrects the source trajectory so that it reconstructs the given recording, then transfers the same correction to the target trajectory (Sec. III-C). Condition decoupling identifies the attributes to change (Sec. III-D, Sec. III-E), and regional preservation assigns stronger source constraints to identity-bearing content (Sec.III-F). These components connect stability under identity with directional change in human-centric editing.

### *C. Residual-Corrected Flow*

Let $z_0$ be the encoded source video and $\epsilon$ a fixed Gaussian sample. We construct a restoration path from this noise to the observed input:

$$V_0 = z_0 - \epsilon, \qquad z_t = (1-t)z_0 + t\epsilon. \tag{3}$$

Following the constant velocity $V_0$ from $t = 1$ to $t = 0$ reconstructs $z_0$. This known path provides a source reconstruction reference at every noise level.

*1) Consistency residual velocity:* At $z_t$, the pretrained model predicts the source-conditioned velocity $V_t^{\text{src}}(z_t) = F_\theta(z_t, t, c_{\text{src}})$. Its difference from the reference velocity defines the consistency residual:

$$V_t^{\text{res}}(z_t) = V_0 - V_t^{\text{src}}(z_t). \tag{4}$$

Adding this residual to the source prediction yields $V_0$, aligning the source-conditioned trajectory with the recorded input, as shown in Fig. 2(b).

*2) Transferring the correction to the target flow:* We use the same residual to correct the target-conditioned velocity, yielding the residual-corrected flow:

$$V_t^{\text{edit}}(z_t^{\text{edit}}) = V_t^{\text{tar}}(z_t^{\text{edit}}) + V_t^{\text{res}}(z_t). \tag{5}$$

Starting from $z_1^{\text{edit}} = \epsilon$, denoising with this field produces the edited video in Fig. 2(c). The target velocity supplies the desired appearance change, while the residual anchors the process to the observed source. Under identical source and target conditions, their common initialization gives coincident trajectories and $V_t^{\text{src}} + V_t^{\text{res}} = V_0$, recovering the source path. When the target condition changes, the target flow introduces the requested change around this reconstruction reference.

For human-centric editing, identity and editable appearance occupy different regions. FlowCine applies the consistency residual to the head mask $H$:

$$V_t^{\text{edit}}(z_t^{\text{edit}}) = V_t^{\text{tar}}(z_t^{\text{edit}}) + H \odot V_t^{\text{res}}(z_t). \tag{6}$$

This concentrates source preservation on facial structure and hair while allowing clothing, props, and the surrounding scene

to follow the target design while it ensures that the purely generative process for new clothing textures and background structures remains free from interference.

### D. Decoupled Conditions

To guide the intended condition difference, we partition the conditioning attributes into an *edit-specific set* $\mathcal{E}$ and an *edit-agnostic set* $\mathcal{U}$ according to the instruction. The former comprises attributes to be modified, such as semantic context, background reference, clothing appearance, object appearance, and illumination, while the latter comprises attributes to be retained, including the subject's identity, structure, and motion. As illustrated in Fig. 3, source and target conditions represent the edit-specific attributes separately, while sharing the conditions that encode edit-agnostic content. This partition operates at the attribute level, that a reference image or text description can contain both editable and preserved information.

*1) Separate appearance conditions:* An image editor converts the instruction $p_{\text{edit}}$ and optional visual assets $\mathcal{A}$ into a target appearance:

$$r_{\text{tar}} = G(r_{\text{src}}, p_{\text{edit}}, \mathcal{A}). \tag{7}$$

Here $r_{\text{src}}$ is the source first frame, and $\mathcal{A}$ may be empty. The source and target images, $r_{\text{src}}$ and $r_{\text{tar}}$, provide distinct appearance references for the original and requested designs. Their associated text descriptions, $q_{\text{src}}$ and $q_{\text{tar}}$, specify the corresponding semantic attributes. Descriptions of preserved content remain common to both, whereas descriptions of edit-specific attributes are updated to reflect the edit. Thus, textual and visual conditions jointly express the intended change without introducing an unnecessary semantic difference in preserved content.

*2) Shared structural conditions:* The same source-derived structural control sequence $S$ is supplied to both branches, providing a common geometric reference as their appearance conditions change. We combine text descriptions, image appearance, and shared source structure as

$$c_{\text{src}} = (q_{\text{src}}, r_{\text{src}}, S), \qquad c_{\text{tar}} = (q_{\text{tar}}, r_{\text{tar}}, S). \tag{8}$$

The target text and image specify the new design, while the common structural condition retains the recorded pose and performance. This decoupling aligns the condition difference with the requested appearance change and supports both text-driven and reference-guided editing through the same interface.

### E. Identity-Aware Regional Structural Control

Structural control follows the same division between preservation and editability. The head receives depth and edge information to retain identity-bearing geometry. The remaining person receives depth to follow the recorded pose while allowing new clothing appearance. The background control is left empty to synthesize freely. With person mask $P$ and head mask $H$, the control is

$$S = H \odot S_{\text{head}} + (P - H) \odot D, \tag{9}$$

where $S_{\text{head}}$ denotes the combination depth and edges by weighted averaging, and $D$ denotes source depth. Figure 4 illustrates this construction.

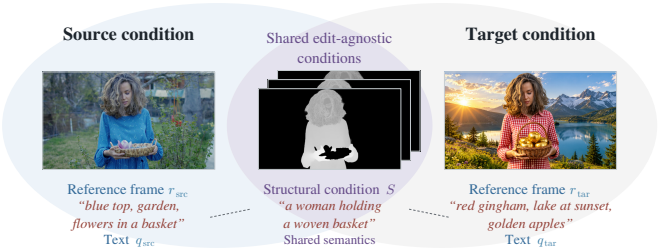



Fig. 3. Decoupled source and target conditions. Edit-specific appearance and text are specified separately for the source and target, while edit-agnostic subject semantics and the source-derived structural sequence $S$ are shared.

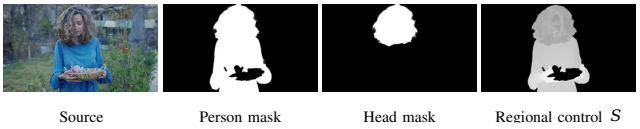



Fig. 4. Source frame, person mask, head mask, and regional control. The head uses mixed depth/edge control, the remaining person uses depth, and the background control is zero.

### F. Fine Detail Preservation using Frequency Transfer

To further improve fine-grained detail fidelity, we introduce high-frequency transfer from the source to the edited video. Let $\text{LF}(u)$ and $\text{HF}(u)$ denote the low- and high-frequency components obtained by Fourier decomposition, with $u = \text{LF}(u) + \text{HF}(u)$. At each denoising time, the source latent $z_t$ supplies fine detail, while the edited latent $z_t^{\text{edit}}$ supplies the evolving appearance. We transfer source high frequencies within the head mask:

$$\begin{aligned} z_t^{\text{edit}} \leftarrow (1 - H) \odot z_t^{\text{edit}} + H \odot \big[ \text{LF}(z_t^{\text{edit}}) \\ + \lambda\, \text{HF}(z_t) + (1 - \lambda)\, \text{HF}(z_t^{\text{edit}}) \big]. \end{aligned} \tag{10}$$

The weight $\lambda \in [0, 1]$ controls the proportion of source high-frequency information. Increasing $\lambda$ strengthens facial and hair detail preservation, while the target's low-frequency component carries the requested appearance. The head mask confines transfer to identity-bearing content, leaving clothing texture and the background under target control.

### G. Acceleration via Residual Reuse

The consistency residual primarily supplies detail compensation that remains relatively stable across neighboring denoising steps, motivating its reuse. To accelerate video editing, we recompute the source-conditioned velocity and its consistency residual every $r$ denoising steps, starting at the first step, and reuse the latest residual between refreshes. The target-conditioned velocity is still evaluated at every step to follow the evolving edited latent. With $r = 10$, our 50-step sampler requires 50 target and only 5 source predictions, corresponding to $1.1\times$ the flow evaluations of pure generation under the same guidance settings. Table VIII evaluates the resulting trade-off between processing time and editing quality.

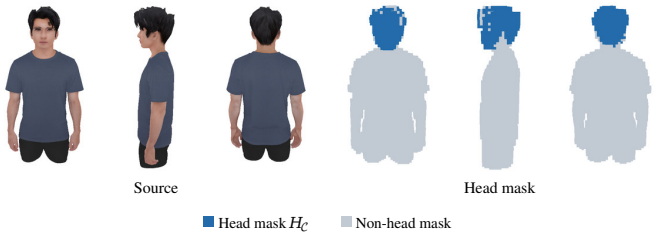



Fig. 5. Spatial head mask for 3D editing. Multiple views show the source asset and its corresponding sparse latent head mask. Blue marks the protected head mask $H_{\mathcal{C}}$; gray marks the non-head mask.

## IV. Residual-Corrected Flow in 3D Latents

### A. Encoding a Supplied Asset

The 3D branch edits an input textured human asset through sparse shape and appearance representations. Encoding produces source features $z_0^{\mathrm{sh}}$ and $z_0^{\mathrm{tex}}$ on a common coarse coordinate set $\mathcal{C}$. The source coordinate set is retained while the feature values change, allowing the decoded asset's geometry and material appearance to follow the requested edit.

### B. Appearance Conditions and a Spatial Head Mask

Source and target views provide appearance conditions $c_{\mathrm{src}}$ and $c_{\mathrm{tar}}$ for the flow models. The image editor constructs the desired appearance from a source view, the edit instruction, and an optional clothing reference. Multi-view source semantic labels are lifted onto the observed surface and associated with sparse latent coordinates, defining a spatial head mask $H_{\mathcal{C}}$. This mask assigns identity preservation consistently across viewing directions while leaving the remaining shape and appearance features editable. Figure 5 shows this spatial mask across views as an example.

### C. Masked Residual-Corrected Flow

The same reconstruction principle applies to both shape and appearance latents. Let $z_0$ denote either source representation and $\epsilon$ shared Gaussian noise. The reference path and its denoising velocity are

$$V_0 = z_0 - \epsilon, \qquad z_t = (1-t)z_0 + t\epsilon. \tag{11}$$

The source- and target-conditioned velocities are denoted by $V_t^{\mathrm{src}}(z_t) = F_\theta(z_t, t, c_{\mathrm{src}})$ and $V_t^{\mathrm{tar}}(z_t^{\mathrm{edit}}) = F_\theta(z_t^{\mathrm{edit}}, t, c_{\mathrm{tar}})$. The consistency residual corrects the source-conditioned prediction toward this known path:

$$V_t^{\mathrm{res}}(z_t) = V_0 - V_t^{\mathrm{src}}(z_t). \tag{12}$$

Using the spatial head mask $H_{\mathcal{C}}$, the edited field becomes

$$V_t^{\mathrm{edit}}(z_t^{\mathrm{edit}}) = V_t^{\mathrm{tar}}(z_t^{\mathrm{edit}}) + H_{\mathcal{C}} \odot V_t^{\mathrm{res}}(z_t). \tag{13}$$

Spatial correction preserves identity-bearing features in the asset's 3D coordinates, while the target condition guides changes in shape and material appearance. Both stages share a source-derived head mask.

### D. Coordinate-Aware Frequency Transfer

As in video editing, we transfer high-frequency information from the source latent $z_t$ to the residual-corrected editing proposal $z_t^{\mathrm{edit}}$ within the spatial head mask $H_{\mathcal{C}}$. Since sparse feature ordering does not reflect spatial proximity, we perform frequency decomposition in 3D coordinate space.

For a sparse latent $u$ on $\mathcal{C}$, we scatter each feature channel onto a grid at its XYZ coordinates, yielding $\widehat{u}$ with zeros at unoccupied locations. The source and edited latents share the same coordinate set $\mathcal{C}$ and binary occupancy mask $O$. Given a Gaussian low-pass operator $L_b$, the occupancy-normalized response at an occupied coordinate is

$$\mathrm{LP}_{b,c}(u) = \left.\frac{L_b(O \odot \widehat{u})}{L_b(O)}\right|_c, \quad c \in \mathcal{C}. \tag{14}$$

Normalizing by filtered occupancy preserves constants at occupied coordinates. Gathering these responses at $\mathcal{C}$ yields the sparse low-pass features $\mathrm{LP}_{b,\mathcal{C}}(u)$. For this coordinate-aware decomposition, the low- and high-frequency components are

$$\begin{aligned} \mathrm{LF}(u) &= \mathrm{LP}_{b,\mathcal{C}}(u), \\ \mathrm{HF}(u) &= u - \mathrm{LF}(u). \end{aligned} \tag{15}$$

We retain the edited low-frequency component and transfer source high frequencies within the head region:

$$\begin{aligned} z_t^{\mathrm{edit}} \leftarrow (1 - H_{\mathcal{C}}) \odot z_t^{\mathrm{edit}} + H_{\mathcal{C}} \odot \big[\mathrm{LF}(z_t^{\mathrm{edit}}) \\ + (1-\eta)\,\mathrm{HF}(z_t) + \eta\,\mathrm{HF}(z_t^{\mathrm{edit}})\big]. \end{aligned} \tag{16}$$

Here $\eta \in [0,1]$ is the retained fraction of edited high-frequency features, while $1-\eta$ controls the contribution from the source. The bandwidth $b$ sets the spatial filtering scale. Smaller $\eta$ strengthens source detail transfer, while the edited low-frequency component carries the requested changes.

### E. Shape-Conditioned Appearance

The shape stage determines the edited geometry, which then conditions appearance editing. Source geometry accompanies the source appearance branch, while edited geometry accompanies the target branch. This coupling adapts the material appearance to the new shape at shared sparse coordinates. Decoding the edited representations produces a textured asset that can be rendered from new views.

## V. Experimental Results

### A. Implementation Details

*1) Evaluation sets:* We constructed a video evaluation set containing 64 video-condition paired cases from 48 distinct video clips with 1-2 condition sets each of 49 frames at different resolutions. Meanwhile, we also conducted a 3D evaluation set contains 24 human assets.

*2) Image and video backbones:* The image editor used is FLUX.2-klein-9B [54]. It receives the source first frame of the video or the front view of the 3D asset, and enabled optional background, clothing, and object references, identified by role in the instruction. The video branch uses Wan2.1-Fun-V1.1-1.3B-Control [55], [56] with 50 steps.

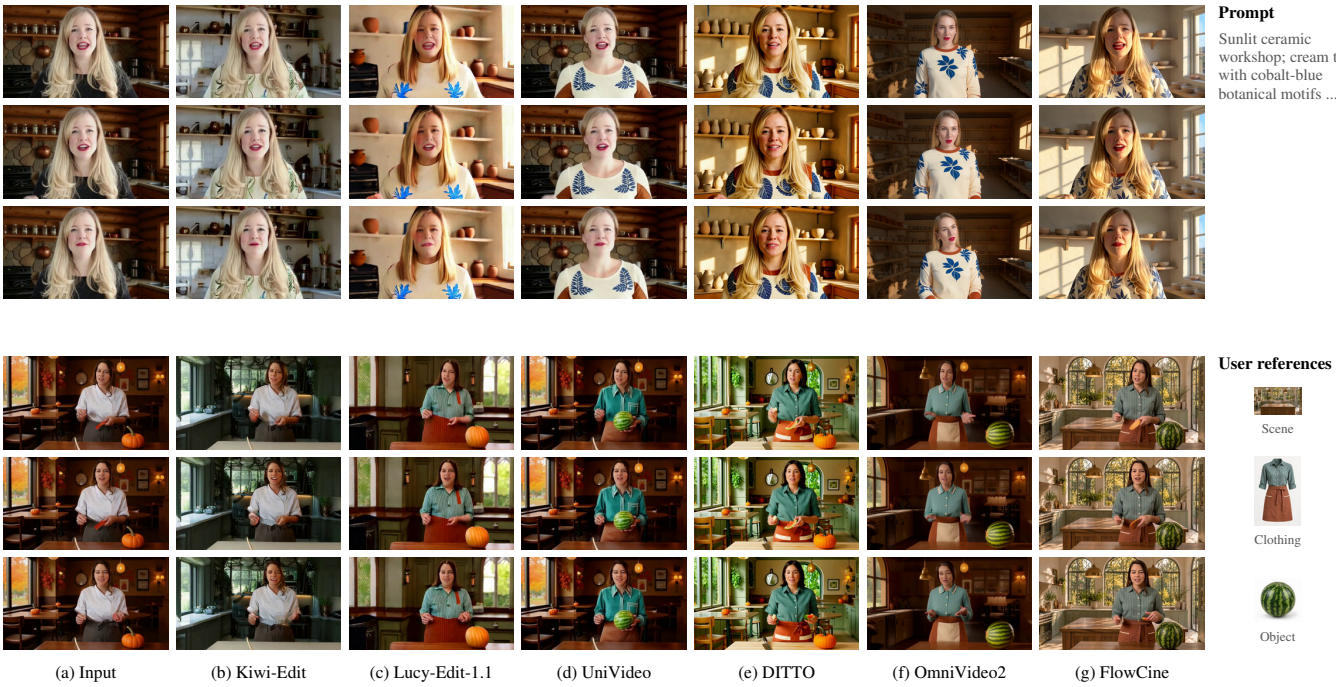



Fig. 6. Qualitative video comparisons. We compare edited sequences under text instructions and visual references, focusing on edit adherence and preservation of the original subject. FlowCine retains facial appearance and expressions while producing coherent changes to scenes, clothing, and objects.

*3) Masks and structural control:* We use FASHN Human Parser [57] to initialize person and face/hair masks. Two independent MatAnyone [58] passes propagate person and head masks. Head control combines depth, Canny, and by weighted average with weights 0.8, 0.1, and 0.1. The remaining person receives depth, and background control is zero.

*4) Video residual correction:* Both source and target CFG scales are set to 6. Frequency transfer uses frequency threshold 0.8 with source high-frequency weight $\lambda = 0.5$ in Eq. (10). To accelerate denoising, we refresh the source prediction every 10 steps and reuse the corresponding residual for the intervening steps. The 50-step process therefore requires 50 target and only 5 source predictions, only $1.1\times$ the flow evaluations of pure generation under the same CFG, while preserving facial identity and fine details.

*5) 3D implementation:* We use TRELLIS.2 [45] at a resolution of 1,024, with 24 sampling steps each for shape and appearance. The source mesh is normalized to the encoder cube and voxelized with material attributes. Shape and appearance features are aligned using exact coordinate keys, with bounded nearest-cell remapping to handle voxelization near source triangle boundaries. We use four source views at 90° azimuth intervals, each at a resolution of $512 \times 768$ with a 35° horizontal field of view. The frontal view is used for image editing, while source and target images share a source-derived crop but use separate person masks. Both flow models condition on images. Source face and hair labels are assigned to latent cells based on the nearest observed surface points. Frequency transfer uses $b = 0.2$ and $\eta = 0.5$. The appearance model concatenates source or edited shape features with the corresponding appearance features. Finally, decoded assets are exported as GLB files in the source coordinate frame, with baked $2048 \times 2048$ PBR textures.

TABLE I
QUANTITATIVE VIDEO EDITING COMPARISON.

| Method | SC ↑ | DC ↑ | MC ↑ | IC ↑ | PA ↑ | AQ ↑ |
|---|---|---|---|---|---|---|
| Kiwi-Edit | 0.7435 | 59.20 | 0.9883 | 0.327 | 0.298 | 5.148 |
| Lucy-Edit-1.1 | <u>0.8126</u> | <u>61.37</u> | 0.9866 | 0.594 | 0.293 | 4.968 |
| UniVideo | 0.7718 | 59.73 | <u>0.9912</u> | <u>0.704</u> | <u>0.304</u> | 4.947 |
| DITTO | 0.6270 | 56.87 | 0.9883 | 0.269 | 0.287 | 5.100 |
| OmniVideo2 | 0.7418 | 58.60 | 0.9532 | 0.333 | 0.303 | <u>5.188</u> |
| **FlowCine(Ours)** | **0.8310** | **61.89** | **0.9934** | **0.719** | **0.305** | **5.325** |

TABLE II
VLM-BASED VIDEO SCORING.

| Method | HC ↑ | IR ↑ | VQ ↑ |
|---|---|---|---|
| Kiwi-Edit | 3.765 | 3.176 | <u>3.647</u> |
| Lucy-Edit-1.1 | <u>3.920</u> | 2.053 | 3.579 |
| UniVideo | 3.864 | <u>3.591</u> | 3.545 |
| DITTO | 2.000 | 1.952 | 3.476 |
| OmniVideo2 | 2.682 | 3.273 | 3.409 |
| **FlowCine(Ours)** | **3.947** | **4.440** | **3.920** |

## B. Evaluation Metrics

We evaluate editing quality and source preservation. Editing quality is assessed in terms of prompt alignment and aesthetics. **Prompt Alignment (PA)** measures CLIP cosine similarity between edited frames and the target prompt [6], and **Aesthetic Quality (AQ)** uses the frozen LAION aesthetic predictor V2 [59]. Source preservation is assessed through identity consistency and regional preservation of structure, detail, and motion. **Identity Consistency (IC)** measures ArcFace feature cosine similarity between corresponding source and edited faces [60]. The regional measures use common source-defined masks. Within the source head mask, **Structure Consistency (SC)** is Canny-map SSIM, while **Detail Consistency (DC)**




Fig. 7. Video module ablation. Removing or altering the components leads to facial detail loss or artifacts in the edited regions. The complete model retains facial appearance and expressions while allowing coherent changes to clothing and surroundings.

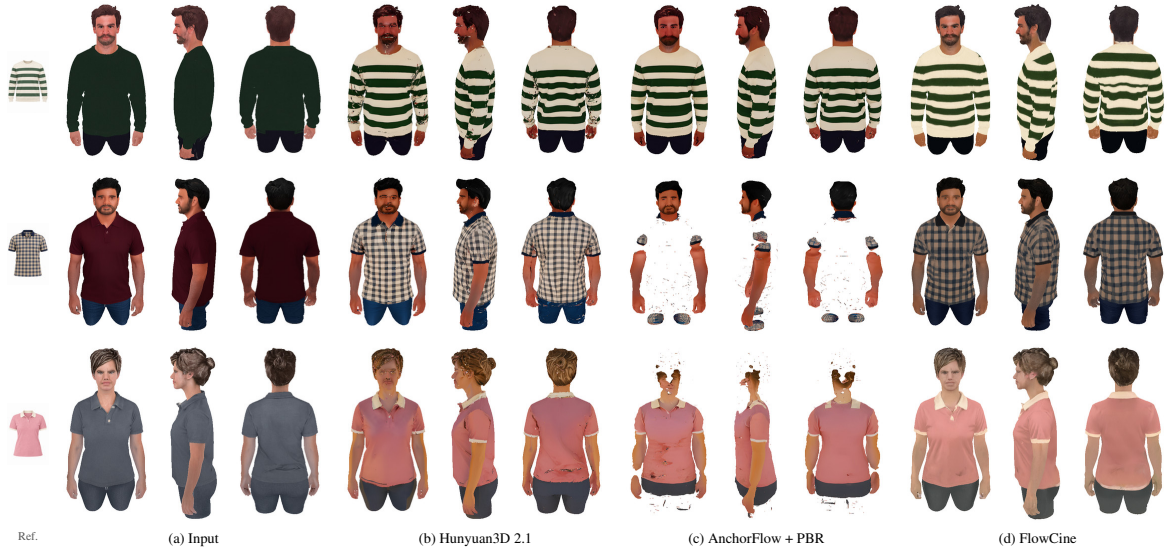



Fig. 8. Qualitative 3D comparisons. Source and edited assets are shown from multiple viewpoints to compare identity preservation and clothing-editing quality. FlowCine retains facial appearance and body structure while producing coherent clothing changes that follow the supplied references.

is PSNR between LoG responses of albedo predictions [61] on the same region, using the frozen Intrinsic v2.0 model for source and output frames. **Motion Consistency (MC)** is SSIM between source and edited RAFT [62] flow components.

We also conducted measures with a vision-language model (VLM) evaluation of video editing. Given aligned source and edited frames, the edit instruction, and supplied reference images, GPT-6.1-sol assesses **Instruction and Reference Following (IR)**, **Video Quality (VQ)**, and **Human Consistency (HC)** which jointly considers structure, fine detail, motion, and identity preservation. The evaluator assigns a score from 1 to 5 for each criterion. Scores are averaged across video editing cases for each criterion.

### C. Video Comparisons

We compare FlowCine with five representative video editing methods that vary in their conditioning interfaces and task coverage. Kiwi-Edit [14] supports both instruction-guided and reference-guided editing, while Lucy Edit 1.1 [15] and DITTO [17] focus on instruction-guided video editing. UniVideo [16] and OmniVideo2 [18] integrate multimodal understanding with video generation and editing. These comparisons examine these methods on human-centric editing, where changes to scenes, clothing, and objects must preserve facial identity, fine details, and recorded motion.

Table I reports the comparison with the above five methods. FlowCine obtains the highest scores on all metrics, indicating the best overall balance between Prompt Alignment,

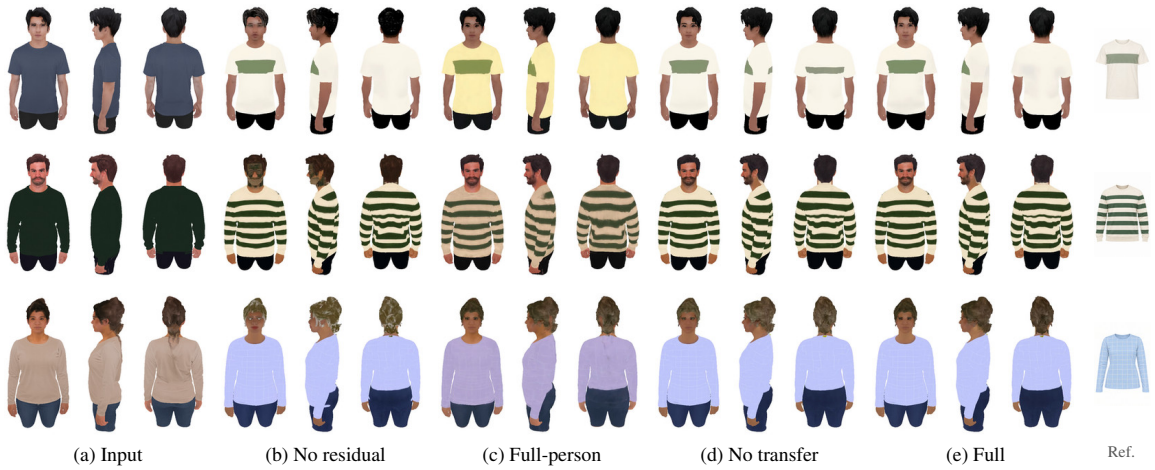



Fig. 9. 3D module ablation. The ablated variants show degradation in facial details or weaker adherence to the target clothing appearance. The full model combines identity preservation with coherent appearance changes across viewpoints.

TABLE III
VIDEO MODULE ABLATION STUDY.

| Variant | SC ↑ | DC ↑ | MC ↑ | IC ↑ | PA ↑ | AQ ↑ |
|---|---|---|---|---|---|---|
| No residual correction | 0.6509 | 57.1364 | 0.9947 | 0.3071 | 0.260 | 5.428 |
| Full video correction | 0.6384 | **59.2576** | **0.9981** | **0.6825** | 0.244 | 5.487 |
| No frequency transfer | 0.6119 | 58.6199 | 0.9971 | 0.6087 | 0.279 | 5.496 |
| **Full** | **0.6580** | 59.1697 | 0.9972 | 0.6705 | **0.282** | **5.506** |

TABLE IV
QUANTITATIVE 3D EDITING COMPARISON.

| Method | SC ↑ | DC ↑ | IC ↑ | PA ↑ | AQ ↑ |
|---|---|---|---|---|---|
| Hunyuan2.1 | 0.707 | 50.234 | 0.402 | 0.313 | 4.603 |
| AnchorFlow + PBR | 0.715 | 47.039 | 0.252 | 0.268 | 4.313 |
| **FlowCine(Ours)** | **0.889** | **60.431** | **0.768** | **0.315** | **4.663** |

TABLE V
3D MODULE ABLATION STUDY.

| Variant | SC ↑ | DC ↑ | IC ↑ | PA ↑ | AQ ↑ |
|---|---|---|---|---|---|
| No residual correction | 0.804 | 54.451 | 0.276 | 0.299 | 4.579 |
| Full-person correction | 0.893 | 59.205 | **0.759** | 0.308 | 4.648 |
| No frequency transfer | 0.883 | 58.037 | 0.645 | 0.305 | 4.701 |
| **Full** | **0.895** | **59.882** | 0.758 | **0.310** | **4.706** |

source preservation, and visual quality. The VLM results in Table II are consistent with this comparison. FlowCine ranks the best in instruction and reference following, video quality, and human consistency. Figure 6 compares the source and method outputs, with the requested prompt and supplied user references alongside each example. The visual comparisons show that FlowCine preserves facial details while applying the requested appearance changes with the best visual quality, supporting the findings of both evaluations. Figure 10 presents additional text-guided and reference-guided video editing results. More video results are available on our project page: https://gaowenshuo.github.io/FlowCine/.

### *D. Video Module Ablation*

We examine three aspects of source preservation: residual correction, its spatial extent, and frequency transfer, while keeping the generation settings fixed. Removing residual correction tests the contribution of compensating for deviations from the source reconstruction path. Applying correction to the full video tests whether this compensation should also constrain regions intended for editing, including clothing and backgrounds. Removing frequency transfer while retaining residual correction isolates the additional contribution of source high-frequency information to fine detail preservation.

The quantitative and qualitative results in Table III and Fig. 7 show that removing residual correction weakens identity preservation, accompanied by changes in facial appearance and expressions. Although full video correction improves several preservation metrics, it reduces prompt alignment and introduces artifacts in edited clothing and backgrounds. This trade-off indicates that spatially unrestricted source constraints can interfere with the intended edits, supporting correction within the head region. Without frequency transfer, facial and hair details deteriorate despite the retained residual correction. These observations support the complementary roles of trajectory correction and detail transfer, with regional restriction maintaining the separation between identity preservation and appearance editing.

### *E. 3D Comparisons*

We compare FlowCine with Hunyuan3D 2.1 [49], which provides shape generation and PBR material synthesis, and AnchorFlow [46], which performs training-free shape editing through latent anchor-aligned flows. Since AnchorFlow operates on geometry, we combine it with Hunyuan PBR painting to obtain textured 3D outputs. This setup evaluates both geometry and appearance, with particular attention to whether requested clothing changes remain consistent across views while preserving source body structure, facial identity, and surface details.

For 3D evaluation, we render the source and edited assets from four fixed viewpoints using identical cameras and unlit base-color rendering. SC is computed on Canny maps, while DC measures the PSNR between LoG responses of linearized




Fig. 10. More video editing results. These examples demonstrate the flexibility of FlowCine across different subjects and various editing conditions.

base-color renders, both within fixed source-defined head masks. IC is evaluated only on valid frontal face pairs. Scores are averaged over valid views within each asset and then equally across assets. Table IV shows that FlowCine achieves the highest scores across all metrics, combining strong structure, detail, and identity preservation with Prompt Alignment and Aesthetic Quality. Figure 8 compares the source and edited assets across viewpoints. FlowCine preserves facial appearance while applying the requested clothing changes consistently across views.

### *F. 3D Module Ablation*

We further examine whether the same preservation mechanisms remain effective in sparse 3D shape and appearance representations, with generation settings fixed. Removing residual correction tests its role in retaining source geometry and appearance during editing. Extending correction from the head region to the full person tests whether broader source constraints benefit identity preservation or interfere with clothing changes. Removing coordinate-aware frequency transfer evaluates whether transferring local source details provides benefits beyond residual correction in the sparse latent space.

Table V shows that removing residual correction degrades structural, detail, and identity consistency, supporting its role in anchoring the edited asset to the source. Full-person correction provides comparable identity preservation to the full model but lower prompt alignment and aesthetic quality, suggesting that broader source constraints offer limited benefit for identity retention while restricting appearance editing. Removing frequency transfer also weakens detail and identity consistency, indicating that residual correction alone is insufficient to fully retain fine surface details. Together with the visual comparisons in Fig. 9, these results support combining correction within the head region with coordinate-aware detail transfer to preserve facial appearance while accommodating the requested clothing changes.

TABLE VI
VIDEO EDITING ACROSS BACKBONES.

| Backbone | SC ↑ | PA ↑ |
|---|---|---|
| Wan2.1 | 0.8310 | 0.305 |
| Wan2.2 | **0.8413** | **0.334** |
| HunyuanVideo | 0.7287 | 0.268 |

### *G. Generalization across Backbones and non-human videos*

We examine backbone generalization for video and 3D editing in Tables VI and VII. The video comparison considers Wan2.1, Wan2.2, and HunyuanVideo, while the 3D comparison considers TRELLIS.2 and Hunyuan3D 2.1. As shown, our method maintains robust performance across different architectures.

TABLE VII
3D EDITING ACROSS BACKBONES.

| Backbone | SC ↑ | PA ↑ |
|---|---|---|
| TRELLIS.2 | **0.889** | **0.315** |
| Hunyuan3D 2.1 | 0.572 | 0.275 |

We further examine FlowCine on videos of non-human subjects. Figure 11 shows scene and illumination edits that preserve the animals' characteristic appearance and motion

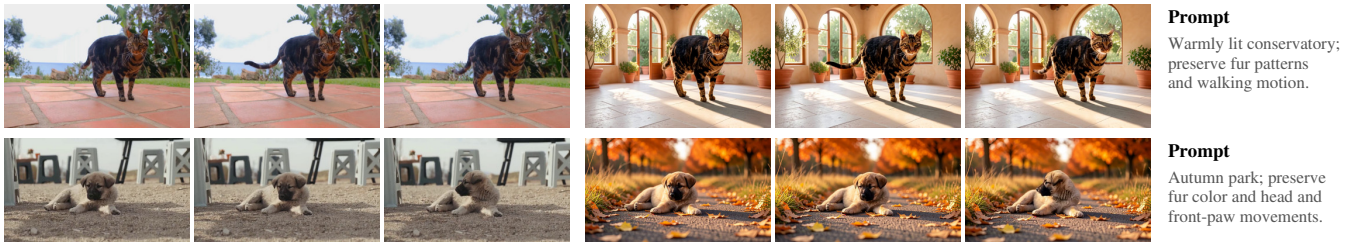



Fig. 11. Video editing on non-human subjects. FlowCine changes the scene and illumination while preserving the animals' appearance and motion.

TABLE VIII
EFFECT OF RESIDUAL REUSING.

| $r$ | DC ↑ | PA ↑ | Time (s) ↓ |
|---|---|---|---|
| 1 | **62.14** | **0.307** | 314 |
| 5 | 61.93 | 0.290 | 195 |
| 10 | 61.89 | 0.305 | 189 |
| 20 | 61.10 | 0.294 | **170** |

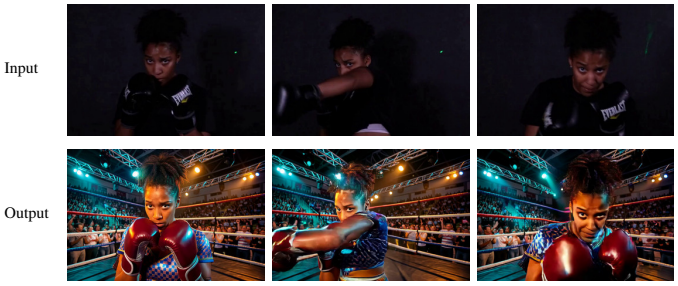



Fig. 12. Limitations on challenging video edits.

across frames. These qualitative results suggest that FlowCine remains robust when applied beyond human-centric scenes.

### H. Analysis of Residual Reusing

We examine the trade-off between editing quality and processing time by varying the residual reuse interval $r$. Table VIII summarizes Detail Consistency, Prompt Alignment, and processing time at different intervals. The results indicate that increasing the reuse step $r$ significantly reduces inference time with minimal impact on quality.

## VI. LIMITATIONS

FlowCine is constrained by the capabilities of the underlying image editing and video generation models. As shown in Fig. 12, it struggles to produce fine details when complex backgrounds, lighting, motion, and clothing are combined.

## VII. CONCLUSION

We presented FlowCine for controllable human-centric editing of videos and 3D assets. Region-dependent structural control, residual correction, and detail transfer coordinate head preservation with scene, clothing, prop, and illumination changes. Text and independently optional visual references provide a common interface. The 3D branch edits shape and PBR latents using semantic spatial masks and coordinate-aware frequency transfer. Video and 3D comparisons examine preservation and requested appearance changes. Future work will develop more generalized editing tasks with more flexible conditions and more self-adaptive structural condition and residual correction mechanism.

## REFERENCES

[1] W. Gao, J. Fan, J. Zeng, and S. Yang, "FlowPortal: Residual-corrected flow for training-free video relighting and background replacement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 2025–2034.
[2] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.
[3] A. Hertz, R. Mokady, J. Tenenbaum, K. Aberman, Y. Pritch, and D. Cohen-Or, "Prompt-to-Prompt image editing with cross-attention control," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.
[4] R. Mokady, A. Hertz, K. Aberman, Y. Pritch, and D. Cohen-Or, "Null-text inversion for editing real images using guided diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 6038–6047.
[5] C. H. Wu and F. De la Torre, "A latent space of stochastic diffusion models for zero-shot image editing and guidance," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 7378–7387.
[6] C. Qi *et al.*, "FateZero: Fusing attentions for zero-shot text-based video editing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 15 932–15 942.
[7] M. Geyer, O. Bar-Tal, S. Bagon, and T. Dekel, "TokenFlow: Consistent diffusion features for consistent video editing," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.
[8] S. Yang, Y. Zhou, Z. Liu, and C. C. Loy, "FRESCO: Spatial-temporal correspondence for zero-shot video translation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 8703–8712.
[9] M. Ku, C. Wei, W. Ren, H. Yang, and W. Chen, "AnyV2V: A tuning-free framework for any video-to-video editing tasks," *Trans. Mach. Learn. Res.*, 2024.
[10] Y. Zhang, X. Ju, and J. J. Clark, "FastVideoEdit: Leveraging consistency models for efficient text-to-video editing," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2025, pp. 3657–3666.
[11] V. Kulikov, M. Kleiner, I. Huberman-Spiegelglas, and T. Michaeli, "FlowEdit: Inversion-free text-based editing using pre-trained flow models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 19 721–19 730.
[12] Y. Bian *et al.*, "VideoPainter: Any-length video inpainting and editing with plug-and-play context control," in *Proc. ACM SIGGRAPH Conf. Papers*, 2025, pp. 1–12.
[13] Z. Jiang, Z. Han, C. Mao, J. Zhang, Y. Pan, and Y. Liu, "VACE: All-in-one video creation and editing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 17 191–17 202.
[14] Y. Lin, G. Liang, Z. Zeng, Z. Bai, Y. Chen, and M. Z. Shou, "Kiwi-Edit: Versatile video editing via instruction and reference guidance," *arXiv preprint arXiv:2603.02175*, 2026.
[15] DecartAI Team, "Lucy Edit: Open-weight text-guided video editing," Technical report; Lucy Edit 1.1 Dev checkpoint. https://huggingface.co/decart-ai/Lucy-Edit-1.1-Dev, 2025, accessed: Oct. 20, 2026.
[16] C. Wei *et al.*, "UniVideo: Unified understanding, generation, and editing for videos," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2026.

[17] Q. Bai *et al.*, "Scaling instruction-based video editing with a high-quality synthetic dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 37 971–37 981.

[18] H. Yang *et al.*, "Omni-Video 2: Scaling MLLM-conditioned diffusion for unified video generation and editing," *arXiv preprint arXiv:2602.08820*, 2026.

[19] X. Cong *et al.*, "VIVA: VLM-guided instruction-based video editing with reward optimization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 34 364–34 374.

[20] J. Mai *et al.*, "EasyV2V: A high-quality instruction-based video editing framework," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 30 435–30 445.

[21] Y. Bai, Y. Gandelsman, M. Gharbi, and Q. Huang, "Instruction-based video editing by repurposing an image editing model," *arXiv preprint arXiv:2608.14790*, 2026.

[22] Y. Wang *et al.*, "VicEdit: Learning to edit videos from visual in-context examples," *arXiv preprint arXiv:2608.16745*, 2026.

[23] A. S. Juvekar *et al.*, "One editor, many edits: A unified training-free framework for diverse video editing," *arXiv preprint arXiv:2609.04190*, 2026.

[24] K. Guo *et al.*, "The relightables: Volumetric performance capture of humans with realistic relighting," *ACM Trans. Graph.*, vol. 38, no. 6, pp. 217:1–217:19, 2019.

[25] L. Zhang, Q. Zhang, M. Wu, J. Yu, and L. Xu, "Neural video portrait relighting in real-time via consistency modeling," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 802–812.

[26] R. Pandey *et al.*, "Total Relighting: learning to relight portraits for background replacement," *ACM Trans. Graph.*, vol. 40, no. 4, pp. 43:1–43:21, 2021.

[27] H. Kim, M. Jang, W. Yoon, J. Lee, D. Na, and S. Woo, "SwitchLight: Co-design of physics-driven architecture and pre-training framework for human portrait relighting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 25 096–25 106.

[28] M. Ren *et al.*, "Relightful Harmonization: Lighting-aware portrait background replacement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 6452–6462.

[29] L. Zhang, A. Rao, and M. Agrawala, "Scaling in-the-wild training for diffusion-based illumination harmonization and editing by imposing consistent light transport," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.

[30] Y. Fang *et al.*, "RelightVid: Temporal-consistent diffusion model for video relighting," *arXiv preprint arXiv:2501.16330*, 2025.

[31] Y. Mei *et al.*, "Lux post facto: Learning portrait performance relighting with conditional video diffusion and a hybrid dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 5510–5522.

[32] Y. Liu *et al.*, "TC-Light: Temporally coherent generative rendering for realistic world transfer," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025, pp. 51 463–51 489.

[33] J. Zeng *et al.*, "Lumen: Consistent video relighting and harmonious background replacement with video generative models," *arXiv preprint arXiv:2508.12945*, 2025.

[34] Y. Lin, Y.-W. Chen, Y.-H. Tsai, R. Clark, and M.-H. Yang, "IllumiCraft: Unified geometry and illumination diffusion for controllable video generation," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025, pp. 27 798–27 829.

[35] K. He *et al.*, "UniRelight: Learning joint decomposition and synthesis for video relighting," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025, pp. 108 435–108 459.

[36] Y. Zhang *et al.*, "LumiSculpt: Enabling consistent portrait lighting in video generation," *arXiv preprint arXiv:2410.22979*, 2024.

[37] P. Liu *et al.*, "UniLumos: Fast and unified image and video relighting with physics-plausible feedback," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025, pp. 82 052–82 080.

[38] W. Gao, X. Lan, and S. Yang, "AnyPortal: Zero-shot consistent video background replacement," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 18 990–18 999.

[39] Y. Zhou *et al.*, "Light-A-Video: Training-free video relighting via progressive light fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 13 315–13 325.

[40] D. Z. Chen, Y. Siddiqui, H.-Y. Lee, S. Tulyakov, and M. Nießner, "Text2Tex: Text-driven texture synthesis via diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 18 558–18 568.

[41] T. Cao, K. Kreis, S. Fidler, N. Sharp, and K. Yin, "TexFusion: Synthesizing 3D textures with text-guided image diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 4169–4181.

[42] A. Haque, M. Tancik, A. A. Efros, A. Holynski, and A. Kanazawa, "Instruct-NeRF2NeRF: Editing 3D scenes with instructions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 19 740–19 750.

[43] Y. Chen *et al.*, "GaussianEditor: Swift and controllable 3D editing with gaussian splatting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 21 476–21 485.

[44] J. Xiang *et al.*, "Structured 3D latents for scalable and versatile 3D generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 21 469–21 480.

[45] ——, "Native and compact structured latents for 3D generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 14 419–14 429.

[46] Z. Zhou *et al.*, "AnchorFlow: Training-free 3D editing via latent anchor-aligned flows," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 14 387–14 397.

[47] M. Parelli, M. Oechsle, M. Niemeyer, F. Tombari, and A. Geiger, "3D-LATTE: Latent space 3D editing from textual instructions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 14 377–14 386.

[48] S. Hu, Y. Wei, F. Zha, Y. Guo, and J. Zhang, "Easy3E: Feed-forward 3D asset editing via rectified voxel flow," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 12 730–12 740.

[49] Team Hunyuan3D *et al.*, "Hunyuan3D 2.1: From images to high-fidelity 3D assets with production-ready PBR material," *arXiv preprint arXiv:2506.15442*, 2025.

[50] Z. Ma, H. Chen, Y. Yue, and G. Gkioxari, "Feedforward 3D editing via text-steerable image-to-3D," *arXiv preprint arXiv:2512.13678*, 2025.

[51] Y. Chi, X. Li, Z. Huang, and J. M. Rehg, "Vinedresser3D: Towards agentic text-guided 3D editing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 12 673–12 683.

[52] E. Sella, H. Phung, N. Amiel, O. Litany, O. Patashnik, and H. Averbuch-Elor, "Prox-E: Fine-grained 3D shape editing via primitive-based abstractions," in *Proc. ACM SIGGRAPH Conf. Papers*, 2026, pp. 1–12.

[53] H. Wen *et al.*, "Learning 3D editing without paired supervision via generative prior distillation," *arXiv preprint arXiv:2609.04942*, 2026.

[54] Black Forest Labs, "FLUX.2 [klein] 9B," Hugging Face model card. https://huggingface.co/black-forest-labs/FLUX.2-klein-9B, 2026, accessed: Oct. 20, 2026.

[55] Team Wan *et al.*, "Wan: Open and advanced large-scale video generative models," *arXiv preprint arXiv:2503.20314*, 2025.

[56] AIGC-Apps, "VideoX-Fun," GitHub repository. https://github.com/aigc-apps/VideoX-Fun, 2025, accessed: Oct. 20, 2026.

[57] FASHN AI, "FASHN human parser: SegFormer for fashion human parsing," Hugging Face model card. https://huggingface.co/fashn-ai/fashn-human-parser, 2024, accessed: Oct. 20, 2026.

[58] P. Yang, S. Zhou, J. Zhao, Q. Tao, and C. C. Loy, "MatAnyone: Stable video matting with consistent memory propagation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 7299–7308.

[59] C. Schuhmann, "LAION aesthetics predictor V2," GitHub repository. https://github.com/christophschuhmann/improved-aesthetic-predictor, 2022, accessed: Oct. 20, 2026.

[60] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 4690–4699.

[61] C. Careaga and Y. Aksoy, "Colorful diffuse intrinsic image decomposition in the wild," *ACM Trans. Graph.*, vol. 43, no. 6, pp. 178:1–178:12, 2024.

[62] Z. Teed and J. Deng, "RAFT: Recurrent all-pairs field transforms for optical flow," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 402–419.